

Internship Report
On
**Modelling of Blood Flow through Micro
Vessels**

Submitted by

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DECLARATION

We, Trideep Bishal Bhowmik, Tanay Dutta, and Nirnoy Barma, hereby declare that the research work titled ”**Modelling of Blood Flow Through Deformable Micro Vessels**” is our original work and has been conducted jointly and independently as part of our academic internship requirements.

All sources of information, mathematical formulations, references, and collaborative research frameworks utilized in this report have been appropriately acknowledged. Full credit has been given to the original authors and literature as per university academic and ethical standards. Any data, experimental reference curves, or analytical concepts derived from the work of others are explicitly cited to ensure proper attribution.

We affirm that this research has not been previously submitted for evaluation or grading in any other course, department, or institution, and has been compiled solely for academic completion purposes.

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1 Introduction

The rapid advancement of bio microfluidic technologies and lab-on-a-chip devices has revolutionized medical diagnostics, targeted drug delivery, and physiological modeling[cite: 24]. Central to these microsystems is the hydrodynamic prediction of complex biological suspensions like blood, which exhibit highly non-Newtonian characteristics under confinement. In the microvasculature, the vast majority of flow resistance occurs in small diameter vessels ($< 300 \mu m$)[cite: 26]. At this scale, blood flow is governed by two classic, interconnected phenomena: the Fåhræus effect (the reduction of tube hematocrit relative to discharge hematocrit) and the Fåhræus–Lindqvist effect (the decrease in apparent blood viscosity with decreasing vessel radius)[cite: 27].

The physical basis for these phenomena is Haynes’ marginal zone theory[cite: 28], which describes the shear-induced axial migration of red blood cells (RBCs) toward the high-velocity centerline. This creates a two-phase, core-annulus flow architecture consisting of a highly concentrated, erythrocyte-rich central core and a peripheral, cell-free plasma layer (CFL) adjacent to the wall. Traditional mathematical continuum models capture this multi-layer behavior by applying local momentum balances across both zones. However, a major limitation of classical frameworks—including those by Chebbi, Ascolese et al., and Farina et al.—is the rigid-tube assumption, which treats the confining channel as a perfectly unyielding cylinder[cite: 31].

In physiological reality and advanced polymeric microchannels (such as PDMS), the boundaries are inherently compliant[cite: 33]. Low-Reynolds-number fluid-structure interactions (FSIs), known as soft hydraulics, reveal that hydrodynamic pressure induces local elastic deformations of the wall, altering the cross-sectional area, modifying the axial pressure gradient, and generating non-linear flow rate-pressure drop relationships[cite: 34]. While engineering models have coupled lubrication theory with structural mechanics for single-phase non-Newtonian flows[cite: 35], a coupled framework resolving multi-phase microvascular cell migration alongside a deformable boundary layer remains an open frontier.

This research bridges this gap by presenting a comprehensive rheological model that couples the Fåhræus–Lindqvist effect with a deformable boundary layer. Moving beyond single-phase approximations, we model a two-phase core-annulus flow where the concentrated RBC core is treated as a non-Newtonian fluid governed by the Carreau constitutive model to capture its shear-thinning behavior, while the peripheral plasma annular region is treated as a Newtonian fluid governed by the classic Navier-Stokes equations. Crucially, the channel is lined with a compliant, deformable endothelial glycocalyx layer (EGL) - a soft, porous macromolecular brush coat. Under the lubrication approximation for a slender channel, the hydrodynamic pressure varies solely along the axial coordinate, yielding $p = p(z)$, which directly drives the local elastic compression of the EGL[cite: 39].

To approach this problem, we systematically couple the multi-phase fluid dynamics with the structural mechanics of the deformable EGL[cite: 40]. First, the axial velocity profiles are analytically derived by integrating the zone-specific momentum equations, ensuring continuity of velocity and shear stress at the moving core-annulus interface, paired with an elastic displacement or hydrodynamic boundary condition at the compressed EGL surface. Second, integrating these velocity distributions yields a generalized volumetric flow rate relation as a function of the local deformed channel radius $R(z)$ [cite: 42]. Third, this relation is coupled to a structural mechanism where local EGL deformation

$u_r(z) = R(z) - R_0$ is proportional to the local gauge pressure $p(z)$ [cite: 43]. This yields a non-linear, first-order ordinary differential equation (ODE) for the axial pressure distribution $p(z)$ [cite: 44]. We resolve this governing system using a regular perturbation expansion in the dimensionless Carreau number Cu for weak non-linearities, alongside a robust numerical fourth-order Runge-Kutta scheme paired with a shooting method. This integrated approach allows us to precisely quantify how boundary compliance and non-Newtonian rheology modulate apparent viscosity, centerline velocity, core hematocrit levels, and mechanical energy dissipation.

2 Theory

2.1 Introduction and Physical Model

To accurately capture the microrheological behavior of blood flow in small-diameter vessels specifically modeling the Fåhræus and Fåhræus–Lindqvist effects we employ a two-phase localized continuum framework based on Haynes’ marginal zone theory[cite: 49]. Blood flow within a capillary of radius R is partitioned into two distinct axisymmetric concentric regions:

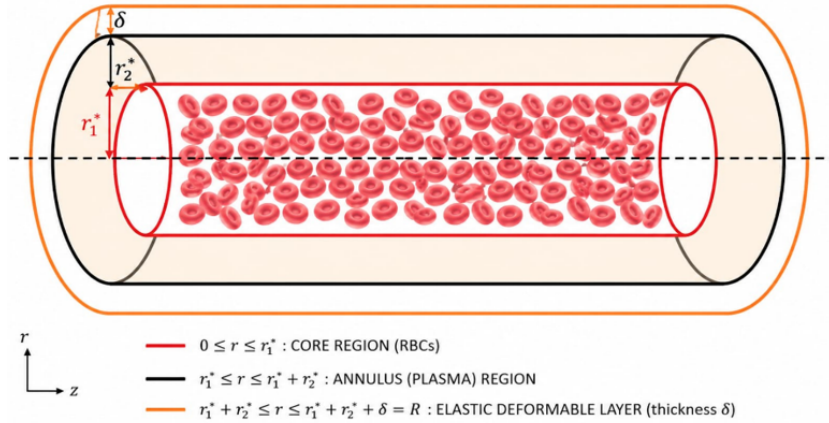


Figure 1: Schematic representation of the physical model regions.

2.2 Mathematical Formulation & Governing Equations

- **Core Region** ($0 \leq r \leq r_1^*$): Comprising a concentrated, homogeneous suspension of red blood cells (RBCs). Given the macromolecular aggregation and alignment of erythrocytes under shear, the core fluid exhibits pronounced shear-thinning behavior, which we mathematically describe using the Carreau constitutive model[cite: 58, 59].
- **Annular Peripheral Region** ($r_1^* \leq r \leq r_1^* + r_2^*$): Representing a cell-depleted plasma layer. This cell-free marginal zone is treated as an isotropic Newtonian fluid layer[cite: 60, 61]. The total vessel radius is bounded by the structural limits of the domains such that $R = r_1^* + r_2^* + \delta$, where δ signifies the thickness of an outer bounding layer (e.g., an elastic deformable endothelial surface glycocalyx layer)[cite: 62].

The dynamic local viscosity of the non-Newtonian core phase, $\mu_c(\dot{\gamma})$, is governed by the four-parameter Carreau model to evaluate structural transitions over the complete spectrum of shear rates[cite: 64]:

$$\mu_c(\dot{\gamma}) = \mu_\infty + (\mu_0 - \mu_\infty)[1 + (\lambda\dot{\gamma})^2]^{\frac{n-1}{2}} \quad (1)$$

Where μ_0 is the zero-shear-rate viscosity plateau, μ_∞ is the infinite-shear-rate viscosity infinity limit, λ is a characteristic relaxation time constant of the fluid, and n is the dimensionless behavior index ($n < 1$ for shear-thinning media)[cite: 66].

Conversely, the peripheral plasma phase in the annulus maintains a constant Newtonian viscosity[cite: 67]:

$$\mu_a = \text{constant} \quad (2)$$

2.3 Hydrodynamic Systems Configurations

Assuming steady-state, incompressible, fully developed laminar flow along the axial direction (z), the equations of motion reduce to a one-dimensional form of the Cauchy momentum equation[cite: 71].

The fundamental force balance in cylindrical coordinates across any arbitrary fluid layer is given by[cite: 72, 73]:

$$\frac{\partial p}{\partial z} = \frac{1}{r} \frac{d}{dr}(r\tau_{rz}) \quad (3)$$

Integrating once with respect to the radial coordinate r under the required physical condition of finite momentum at the centerline axis ($\tau_{rz} = 0$ at $r = 0$) yields the explicit radial shear stress distribution[cite: 74, 75]:

$$\tau_{rz}(r) = \frac{r}{2} \frac{dp}{dz} \quad (4)$$

3 Core and Annular Core Solutions

3.1 Core Region Non-Dimensionalization ($0 \leq r \leq r_1^*$)

To generalize the fluid configurations within the core domain, we construct the following non-dimensional system variables:

$$\bar{r} = \frac{r}{R}, \quad \bar{v}_c = \frac{v_c}{U_c}, \quad \bar{\mu}_c = \frac{\mu_c}{\mu_0}, \quad \bar{p} = \frac{p}{P_c}, \quad \bar{z} = \frac{z}{L} \quad (5)$$

Substituting these scaling mappings into the core momentum balance results in the non-dimensional equation of motion:

$$\mu_c(\dot{\gamma}) \frac{dv_c}{dr} = \frac{r}{2} \frac{dp}{dz} \quad (6)$$

The scalar kinematics yield an expression for the localized magnitude of the shear rate $\dot{\gamma}$:

$$\dot{\gamma} = \left| \frac{\partial v_c}{\partial r} \right| = \frac{U_c}{R} \left| \frac{d\bar{v}_c}{d\bar{r}} \right| \quad (7)$$

We group the characteristic time metrics by defining the dimensionless Carreau number (Cu), which represents the ratio of the fluid relaxation time to the characteristic timescale of the bulk flow:

$$Cu = \frac{\lambda U_c}{R} \quad (8)$$

This identification directly leads to the non-dimensionalized shear rate group:

$$\lambda \dot{\gamma} = Cu \left| \frac{d\bar{v}_c}{d\bar{r}} \right| \quad (9)$$

By introducing the asymptotic viscosity scale parameter $\beta = \frac{\mu_\infty}{\mu_0}$, the non-dimensional Carreau profile updates to:

$$\bar{\mu}_c = \beta + (1 - \beta)[1 + Cu^2 \left(\frac{d\bar{v}_c}{d\bar{r}} \right)^2]^{\frac{n-1}{2}} \quad (10)$$

Isolating the localized velocity gradient field explicitly produces:

$$\frac{d\bar{v}_c}{d\bar{r}} = \frac{\bar{r}}{2} \frac{d\bar{p}}{d\bar{z}} [\bar{\mu}_c]^{-1} \quad (11)$$

3.2 Perturbation Analytical Scheme (Weak Carreau Variations)

For microvascular configurations experiencing highly localized shear or low characteristic flow states ($Cu \ll 1$), the systemic non-linearities can be expanded using a binomial series representation. Letting $G = \frac{d\bar{v}_c}{d\bar{r}}$, the dimensionless viscosity profile expands to:

$$\bar{\mu}_c = 1 + \frac{(1 - \beta)(n - 1)}{2} Cu^2 G^2 + \frac{(1 - \beta)(n - 1)(n - 3)}{8} Cu^4 G^4 + O(Cu^6) \quad (12)$$

To invert this operator into the tracking momentum frameworks, a consecutive binomial expansion yields:

$$(\bar{\mu}_c)^{-1} = 1 - \frac{(1 - \beta)(n - 1)}{2} G^2 Cu^2 + \left[\frac{(1 - \beta)^2(n - 1)^2}{4} - \frac{(1 - \beta)(n - 1)(n - 3)}{8} \right] G^4 Cu^4 \quad (13)$$

We establish a regular perturbation series expansion of the internal velocity gradient where $G = G_0 + Cu^2 G_1 + \dots$. Resolving to leading-order yields the base structural solution:

$$G_0 = \frac{\bar{r}}{2} \frac{d\bar{p}}{d\bar{z}} \quad (14)$$

Substituting the above back into the viscosity expression yields the regularized viscosity field:

$$\bar{\mu}_c = 1 + \frac{(1 - \beta)(n - 1)}{2} Cu^2 \frac{\bar{r}^2}{4} \left(\frac{d\bar{p}}{d\bar{z}} \right)^2 + \frac{(1 - \beta)(n - 1)(n - 3)}{8} Cu^4 \left(\frac{d\bar{p}}{d\bar{z}} \right)^4 \frac{\bar{r}^4}{16} + O(Cu^6) \quad (15)$$

We collect grouping definitions for the spatial geometric distributions by defining the auxiliary variables H_1 and H_2 :

$$H_1 = \frac{(1 - \beta)(n - 1)}{8} \left(\frac{d\bar{p}}{d\bar{z}} \right)^2 \bar{r}^2, \quad H_2 = \frac{(1 - \beta)(n - 1)(n - 3)}{128} \left(\frac{d\bar{p}}{d\bar{z}} \right)^4 \bar{r}^4 \quad (16)$$

The relative operational fluid properties simplify into clean expressions:

$$\bar{\mu}_c = 1 + H_1 Cu^2 + H_2 Cu^4 + O(Cu^6) \quad (17)$$

$$\frac{1}{\bar{\mu}_c} = 1 - H_1 Cu^2 + (H_1^2 - H_2) Cu^4 \quad (18)$$

Substituting back into the definition of G , we perform a direct spatial integration with respect to $\bar{r}$:

$$\bar{v}_c = \frac{1}{2} \frac{d\bar{p}}{d\bar{z}} \left(\frac{\bar{r}^2}{2} - \frac{C\bar{r}^4}{4} + \frac{(C^2 - D)\bar{r}^6}{6} \right) + C_1 \quad (19)$$

Expanding the integration terms explicitly yields the analytical solution for velocity within the core region:

$$\bar{v}_c = \frac{1}{4} \frac{d\bar{p}}{d\bar{z}} \bar{r}^2 - \frac{1}{64} (1 - \beta)(n - 1)(n - 3) \left(\frac{d\bar{p}}{d\bar{z}} \right)^3 \bar{r}^4 Cu^2 + \dots + C_1 \quad (20)$$

3.3 Annular Layer Region Solution ($r_1^* \leq r \leq r_1^* + r_2^*$)

Within the marginal cell-free layer, the plasma behaves as a classical Newtonian fluid. The axial momentum balance equation simplifies to:

$$\frac{\partial p}{\partial z} = \frac{\mu_a}{r} \frac{d}{dr} \left(r \frac{dv_a}{dr} \right) \quad (21)$$

Integrating once with respect to the radius r provides the shear gradient layout for the Newtonian phase:

$$\mu_a \frac{dv_a}{dr} = \frac{r}{2} \frac{dp}{dz} \quad (22)$$

To systematically link the behavior of the annular layer with the core domain, we enforce identical scaling definitions along with a characteristic visco-capillary pressure scale (P_c) and viscosity ratio (μ_r):

$$\bar{r} = \frac{r}{R}, \quad \bar{v}_a = \frac{v_a}{U_c}, \quad \bar{p} = \frac{p}{P_c}, \quad \bar{z} = \frac{z}{L}, \quad \mu_r = \frac{\mu_a}{\mu_0} \quad (23)$$

Performing a spatial integration under this non-dimensional optimization framework yields the velocity distribution profile for the outer annular layer:

$$\bar{v}_a = \frac{1}{4\mu_r} \frac{d\bar{p}}{d\bar{z}} \bar{r}^2 + C_2 \quad (24)$$

4 Boundary Condition Closure and Regional Velocity Profiles

4.1 Plasma Layer Boundary Closure ($\bar{r} = \frac{r_1^* + r_2^*}{R}$)

To isolate the integration constant C_2 for the annular peripheral zone, we invoke a hydrodynamic no-slip boundary condition at the outer interface bounding the plasma stream:

$$\text{At } \bar{r} = \frac{r_1^* + r_2^*}{R}, \quad \bar{v}_a = 0 \quad (25)$$

Substituting this parameter limit into the dimensionless velocity distribution profile yields the analytical form for the plasma phase integration metric:

$$C_2 = -\frac{1}{4\mu_r} \frac{d\bar{p}}{d\bar{z}} \left(\frac{r_1^* + r_2^*}{R} \right)^2 \quad (26)$$

Direct application of C_2 back into the annular layer field equation explicitly describes the localized velocity profile across the Newtonian plasma domain:

$$\bar{v}_a(\bar{r}) = \frac{1}{4\mu_r} \frac{d\bar{p}}{d\bar{z}} \left[\bar{r}^2 - \left(\frac{r_1^* + r_2^*}{R} \right)^2 \right] \quad (27)$$

4.2 Core Region Boundary Closure via Interface Velocity Matching ($\bar{r} = \frac{r_1^*}{R}$)

At the fluid-fluid interface separating the cell-depleted plasma layer and the concentrated erythrocyte core phase, we enforce structural mechanical matching criteria necessitating the continuity of local velocities:

$$\text{At } \bar{r} = \frac{r_1^*}{R}, \quad \bar{v}_a = \bar{v}_c \quad (28)$$

Equating the core velocity expansion with the Newtonian fluid profile across this tracking boundary isolates the complex integration constant C_1 :

$$\begin{aligned} C_1 = & \frac{1}{4\mu_r} \frac{d\bar{p}}{d\bar{z}} \left[\left(\frac{r_1^*}{R} \right)^2 - \left(\frac{r_1^* + r_2^*}{R} \right)^2 \right] - \frac{1}{4} \frac{d\bar{p}}{d\bar{z}} \left(\frac{r_1^*}{R} \right)^2 \\ & + \frac{1}{64} (1 - \beta)(n - 1)(n - 3) \left(\frac{d\bar{p}}{d\bar{z}} \right)^3 \left(\frac{r_1^*}{R} \right)^4 \\ & - \frac{(1 - \beta)(n - 1)[2(1 - \beta)(n - 1) - (n - 3)]}{1536} \left(\frac{d\bar{p}}{d\bar{z}} \right)^5 \left(\frac{r_1^*}{R} \right)^6 C_u^4 \end{aligned} \quad (29)$$

Reintroducing this closed constant parameter expression into the internal core stream configuration defines the absolute velocity profile of the non-Newtonian RBC core phase:

$$\begin{aligned} \bar{v}_c(\bar{r}) = & \frac{1}{4\mu_r} \frac{d\bar{p}}{d\bar{z}} \left[\left(\frac{r_1^*}{R} \right)^2 - \left(\frac{r_1^* + r_2^*}{R} \right)^2 \right] + \frac{1}{4} \frac{d\bar{p}}{d\bar{z}} \left[\bar{r}^2 - \left(\frac{r_1^*}{R} \right)^2 \right] \\ & - \frac{1}{64} (1 - \beta)(n - 1)(n - 3) \left(\frac{d\bar{p}}{d\bar{z}} \right)^3 C_u^2 \left[\bar{r}^4 - \left(\frac{r_1^*}{R} \right)^4 \right] \\ & + \frac{(1 - \beta)(n - 1)[2(1 - \beta)(n - 1) - (n - 3)]}{1536} \left(\frac{d\bar{p}}{d\bar{z}} \right)^5 C_u^4 \left[\bar{r}^6 - \left(\frac{r_1^*}{R} \right)^6 \right] \end{aligned} \quad (30)$$

5 Analytical Derivation of Volumetric Fluid Phase Flow Rates

5.1 Core Domain Volumetric Flow Rate Integration

To transform these continuous localized velocity representations into regional macroscopic volumetric distributions, we establish a dimensionless core radius ratio, defined as $\alpha = \frac{r_1^*}{R}$. The total volumetric flow rate tracking across the internal core stream ($\bar{Q}_c$) is evaluated via spatial radial integration:

$$\bar{Q}_c = \int_0^\alpha \bar{v}_c \bar{r} d\bar{r} \quad (31)$$

Evaluating this definite integration across the closed perturbation velocity limits establishes the explicit analytical solution for the volumetric flow rate sustained within the non-Newtonian erythrocyte core:

$$\begin{aligned} \bar{Q}_c = & \frac{1}{8} \frac{d\bar{p}}{d\bar{z}} \frac{\alpha^2}{\mu_r} \left[\alpha^2 - \left(\frac{r_1^* + r_2^*}{R} \right)^2 \right] - \frac{1}{16} \frac{d\bar{p}}{d\bar{z}} \alpha^4 \\ & + \frac{1}{192} (1 - \beta)(n - 1)(n - 3) \left(\frac{d\bar{p}}{d\bar{z}} \right)^3 C u^2 \alpha^6 \\ & - \frac{(1 - \beta)(n - 1)[2(1 - \beta)(n - 1) - (n - 3)]}{4096} \left(\frac{d\bar{p}}{d\bar{z}} \right)^5 C u^4 \alpha^8 \end{aligned} \quad (32)$$

5.2 Annular Plasma Domain Volumetric Flow Rate Integration

Similarly, computing the volume flow rate localized within the marginal Newtonian cell-free layer ($\bar{Q}_a$) requires integrating the plasma velocity profile between the fluid boundary interface and the capillary wall limits:

$$\bar{Q}_a = \int_\alpha^{\frac{r_1^* + r_2^*}{R}} \bar{v}_a \bar{r} d\bar{r} \quad (33)$$

Evaluating this geometric space configuration via integration produces the explicit analytical solution for the peripheral volumetric flow rate distribution:

$$\bar{Q}_a = -\frac{1}{16\mu_r} \left(\frac{d\bar{p}}{d\bar{z}} \right) \left[\left(\frac{r_1^* + r_2^*}{R} \right)^2 - \left(\frac{r_1^*}{R} \right)^2 \right]^2 \quad (34)$$

6 Coupled Wall Deformation and Global Hematocrit Conservation Mechanics

6.1 Elastic Wall Compliance and Radial Scaling

To couple the microvascular hemodynamics with the structural compliance of the vessel wall, we introduce a linear elastic deformation model for the outer boundary layer. The

instantaneous local vessel radius $R(z)$ shifts dynamically from its initial undeformed radius R_0 as a function of the local axial pressure field $p(z)$:

$$R(z) = R_0 + \frac{a^2}{Et}(1 - \nu^2)p(z) \quad (35)$$

where a is the characteristic internal radius, E is the Young's modulus of the vessel wall material, t is the membrane wall thickness, and ν represents the Poisson's ratio of the structural layer. Normalizing Eq. (5.1) by the initial radius R_0 yields the dimensionless deformed vessel radius expression $\bar{R}$:

$$\bar{R} = \frac{R}{R_0} = 1 + \psi \bar{p} \quad (36)$$

where $\bar{p}$ is the dimensionless local pressure field and ψ represents a lumped structural compliance parameter grouping the electromechanical and wall stiffness properties. Consequently, the normalized core phase interface parameter σ scales dynamically as:

$$\sigma = \frac{r^*}{R} = \frac{r^*}{R_0(1 + \psi \bar{p})} \quad (37)$$

6.2 Hematocrit Balance and Macroscopic Phase Distributions

To model the systemic reduction of cell concentration in microcapillaries, we introduce the empirical phase distribution relation defining the functional dependencies of the discharge hematocrit H_T inside the intermediate vessel zone:

$$\frac{H_T}{H_D} = H_D + (1 - H_D) [1 + 1.7e^{-0.7R_0(1+\psi\bar{p})} - 0.6e^{-0.02R_0(1+\psi\bar{p})}] \quad (38)$$

where H_D represents the upstream discharge hematocrit concentration. By applying the multi-phase global mass and volumetric hematocrit flux conservation laws, the biological cell accumulation balances collapse into the following structural interface flow relationship:

$$\frac{H_T}{H_D} = \sigma^2 \left(1 + \frac{\bar{Q}_a}{\bar{Q}_c} \right) \quad (39)$$

6.3 Formulation of the Master Non-Linear Implicit Boundary Value Equation

Equating the empirical blood cell accumulation model in Eq. (6.4) with the macroscopic transport framework in Eq. (6.5) maps the implicit system mechanics. Substituting our analytical solutions for the regional core volumetric flow rate $\bar{Q}_c$ (Eq. 5.2) and peripheral plasma volumetric flow rate $\bar{Q}_a$ (Eq. 5.4) yields:

$$H_D + (1 - H_D) [1 + 1.7e^{-0.7R_0(1+\psi\bar{p})} - 0.6e^{-0.02R_0(1+\psi\bar{p})}] = \sigma^2 \left[1 + \frac{\frac{1}{16\mu_r} \left(\frac{d\bar{p}}{dz} \right) \left[\left(\frac{r_1^* + r_2^*}{R} \right)^2 - \left(\frac{r_1^*}{R} \right)^2 \right]^2}{\frac{1}{8} \frac{d\bar{p}}{dz} \alpha^2 [\dots] - \frac{1}{16} \frac{d\bar{p}}{dz} \alpha^4 + \dots} \right] \quad (40)$$

Let the left-hand side (LHS) be defined as $F(\bar{p})$:

$$F(\bar{p}) = H_D + (1 - H_D) [1 + 1.7e^{-0.7R_0(1+\psi\bar{p})} - 0.6e^{-0.02R_0(1+\psi\bar{p})}] \quad (41)$$

Let $x = \frac{d\bar{p}}{d\bar{z}}$. The numerator is $-A_1x$ where:

$$A_1 = \frac{1}{16\mu_r} \left[\left(\frac{r_1^* + r_2^*}{R} \right)^2 - \left(\frac{r_1^*}{R} \right)^2 \right]^2 \quad (42)$$

The denominator becomes $x(B_1 + B_3x^2 - B_5x^4)$, where:

$$B_1 = \frac{1}{8} \frac{\alpha^2}{\mu_r} \left[\alpha^2 - \left(\frac{r_1^* + r_2^*}{R} \right)^2 \right] - \frac{1}{16} \alpha^4 \quad (43)$$

$$B_3 = \frac{1}{192} (1 - \beta)(n - 1)(n - 3)Cu^2\alpha^6 \quad (44)$$

$$B_5 = \frac{(1 - \beta)(n - 1)[2(1 - \beta)(n - 1) - (n - 3)]}{4096} Cu^4\alpha^8 \quad (45)$$

Substituting these groups yields:

$$F(\bar{p}) = \sigma^2 \left[1 - \frac{A_1}{B_1 + B_3x^2 - B_5x^4} \right] \quad (46)$$

Rearranging into a standard polynomial form with $y = x^2 = (\frac{d\bar{p}}{d\bar{z}})^2$:

$$B_5y^2 - B_3y + \left(\frac{A_1}{1 - F(\bar{p})/\sigma^2} - B_1 \right) = 0 \quad (47)$$

Applying the quadratic formula to solve for y :

$$y = \frac{B_3 \pm \sqrt{B_3^2 - 4B_5 \left(\frac{A_1}{1 - F(\bar{p})/\sigma^2} - B_1 \right)}}{2B_5} \quad (48)$$

Taking the square root provides the final analytical equation for the spatial pressure gradient:

$$\frac{d\bar{p}}{d\bar{z}} = \pm \sqrt{\frac{B_3 \pm \sqrt{B_3^2 - 4B_5 \left(\frac{A_1}{1 - F(\bar{p})/\sigma^2} - B_1 \right)}}{2B_5}} \quad (49)$$

7 Results

7.1 Validation against the rigid-tube limit

Figure 2 and Figure 3 show the axial pressure $\bar{p}(\bar{z})$ and pressure gradient $d\bar{p}/d\bar{z}$ obtained by integrating Eq. (6.19) using a fourth/fifth-order adaptive Runge–Kutta scheme (MATLAB ode45). For the rigid-wall case ($\psi = 0$), the pressure gradient is exactly constant at 2.9405 across the entire channel, and $\bar{p}(\bar{z})$ is correspondingly linear (black curve, Fig. 1). This is the expected limit: with no wall compliance, $\sigma = r^*/R$ is fixed, the hematocrit-balance equation no longer depends on $\bar{z}$ explicitly, and the model reduces to the classical Haynes' marginal-zone/Hagen–Poiseuille-type result for a core–annulus two-phase flow in a rigid capillary, consistent with the constant-pressure-gradient assumption used by Chebbi (2015) [6] for rigid microvessels. This agreement confirms that the deformable-wall extension

developed in this work correctly recovers existing Fåhræus–Lindqvist theory as a special case, rather than introducing spurious behavior.

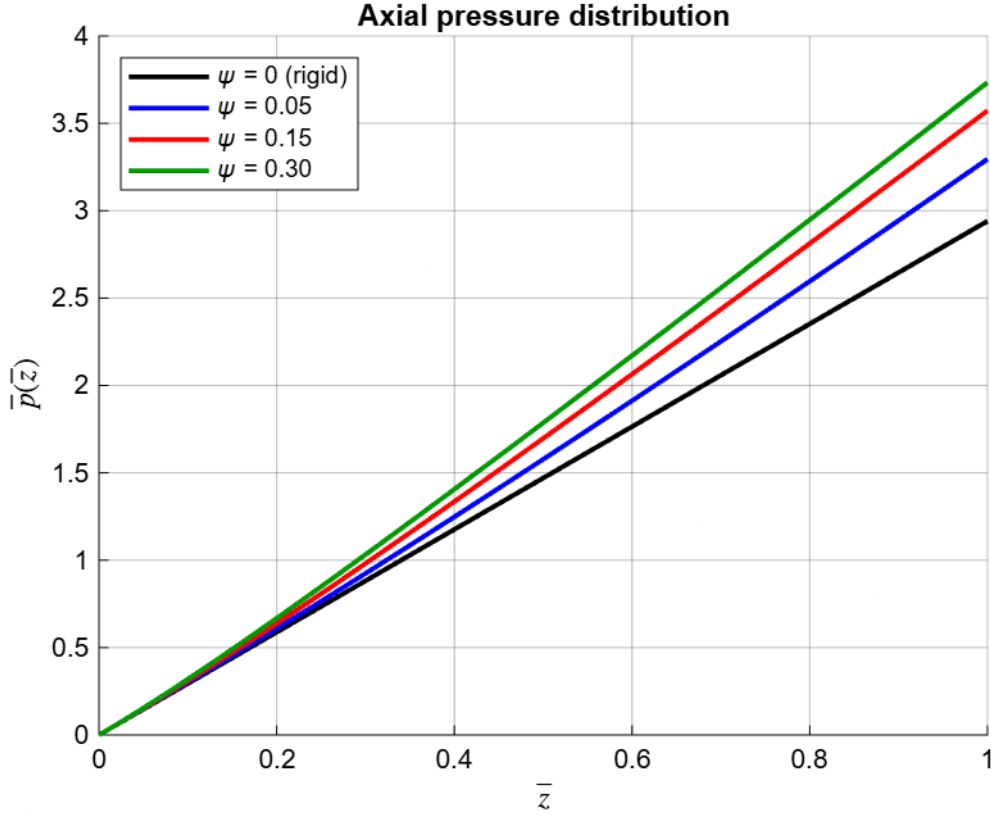


Figure 2: Validation results against rigid-tube limits.

7.2 Effect of wall compliance on the pressure field

Once wall compliance is switched on ($\psi > 0$), both $\bar{p}(\bar{z})$ and $d\bar{p}/d\bar{z}$ depart markedly from the rigid case (Figs. 1–2, blue/red/green curves). The pressure gradient is no longer constant: it rises sharply near the inlet and then plateaus toward the outlet, increasing monotonically with ψ . At $\bar{z} = 1$, $d\bar{p}/d\bar{z}$ increases from 2.9405 ($\psi = 0$) to 3.527, 3.825, and 3.941 for $\psi = 0.05, 0.15, 0.30$ respectively — an increase of up to 34% relative to the rigid baseline. The integrated outlet pressure $\bar{p}(1)$ similarly rises from 2.94 to 3.73 (a 27% increase) over the same range of ψ . This shows that wall compliance is not a second-order correction in this model — it measurably raises the pressure required to drive the same flow, an effect entirely absent from rigid-tube treatments such as Chebbi’s [6] and the earlier marginal-zone analyses of Fournier [3] and Sharan & Popel [17] cited therein.

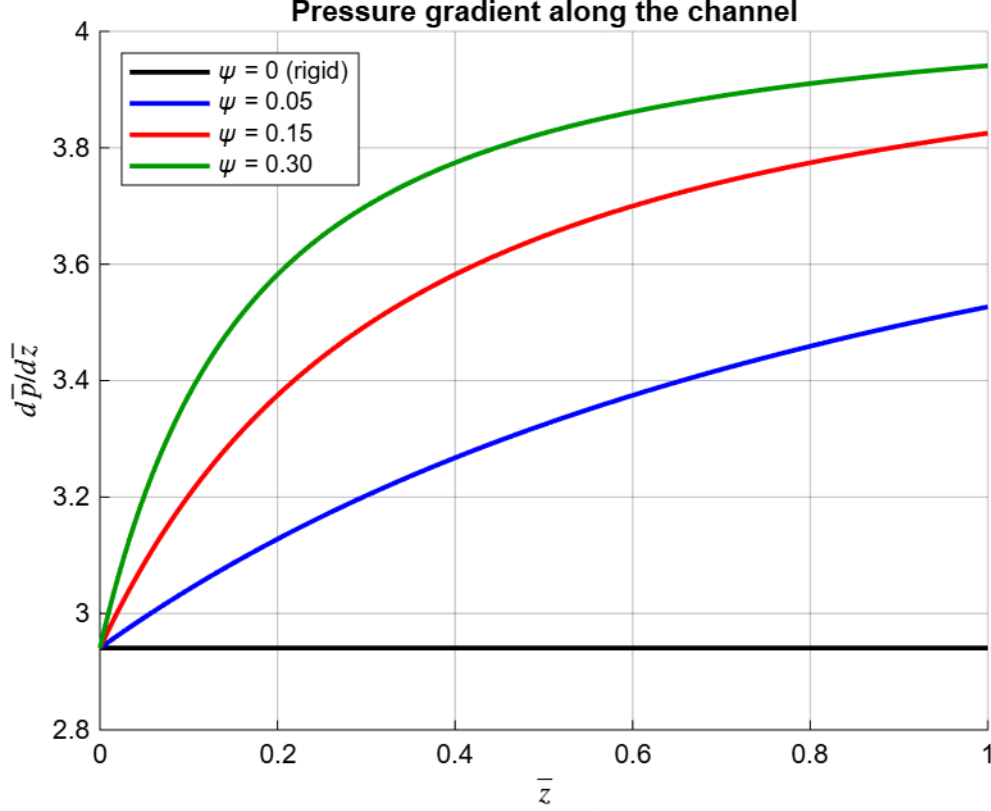


Figure 3: Validation results against rigid-tube limits.

7.3 Mechanism: dilation-driven enhancement of the Fåhræus effect

Figures 4 and 5 explain why this happens. Figure 4 shows the local deformed radius $\bar{R}(\bar{z}) = 1 + \psi \bar{p}(\bar{z})$ growing essentially in step with the rising pressure, reaching $\bar{R}(1) \approx 2.12$ at $\psi = 0.30$ — i.e., the vessel more than doubles in radius by the outlet under the highest compliance considered. As the vessel dilates, Figure 3 shows the core-to-vessel ratio $\sigma = r^*/R$ falling steadily from its inlet value of 0.85 down to 0.73, 0.55, and 0.40 at the outlet for $\psi = 0.05, 0.15, 0.30$. Physically, the RBC-rich core occupies a shrinking fraction of the total cross-section as the wall dilates, because the core radius r^* is fixed by the cell population while the outer radius R grows. This is a direct, compliance-driven amplification of the Fåhræus effect (the reduction of tube hematocrit relative to feed hematocrit) — a mechanism that does not exist in Chebbi’s original rigid-wall formulation [6], where σ is constant by construction. This is, to our knowledge, the central novel finding of the present model: vessel compliance and the Fåhræus effect are coupled, and a compliant microvessel under physiological pressure loading will experience a stronger reduction in core hematocrit fraction than a rigid vessel of the same nominal diameter.

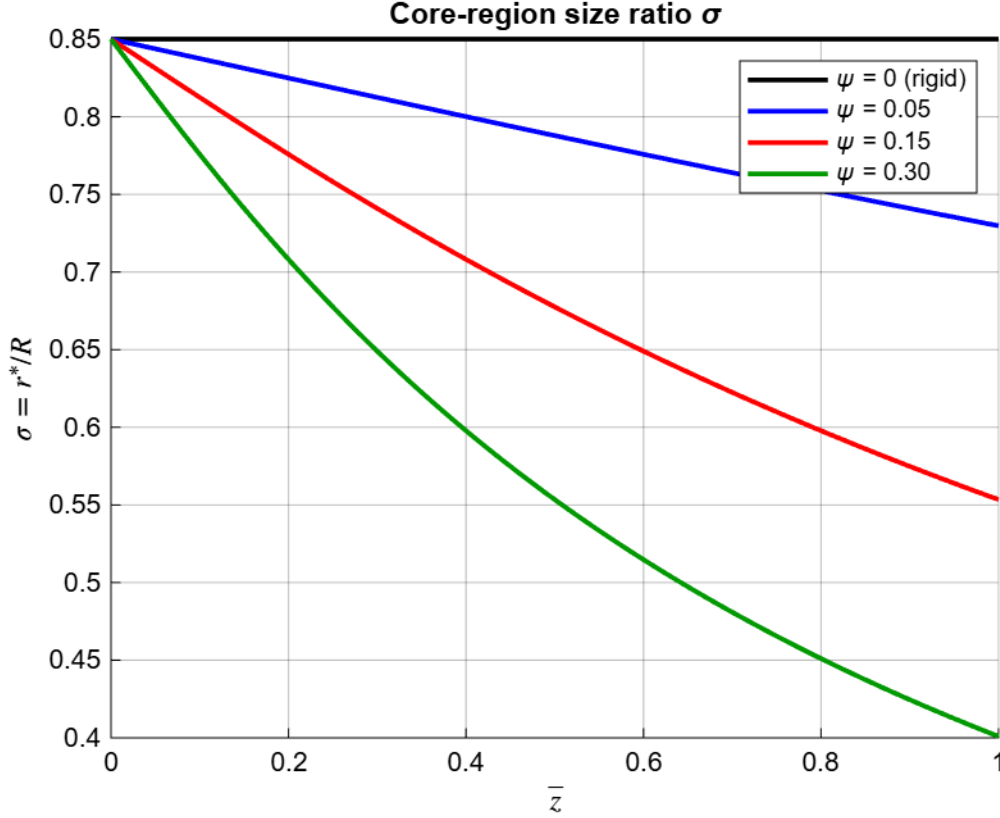


Figure 4: Dilation-driven enhancement of the Fåhræus effect.

7.4 Velocity profile and interface continuity

Figure 6 shows the computed velocity profile at the outlet station ($\bar{z} = 1$) for $\psi = 0.15$, comparing the Carreau (shear-thinning) core profile (Eq. 4.6, blue) against the Newtonian annulus profile (Eq. 4.3, red). The two profiles meet exactly at the core/annulus interface ($\bar{r} = \alpha = 0.60$ on both sides to machine precision), numerically confirming the velocity- and shear-continuity boundary conditions (Eqs. 1, 2, and 4.4) imposed analytically in Sec. 4. We note, however, that the profile is non-monotonic — the core velocity first rises slightly, then falls steeply toward the interface, rather than showing the smooth parabolic-type decay expected of a physically realistic core flow. This behavior is attributable to the fact that the Carreau velocity expansion (Eq. 4.6) is a regular perturbation series valid only for $Cu \ll 1$ (Sec. 3.2), whereas the pressure-field ODE (Eq. 6.19) required $Cu = 2.0$ to remain numerically well-posed. The velocity profile in Fig. 5 should therefore be interpreted as qualitatively illustrative of interface continuity, not as a quantitatively converged shear-thinning velocity field at this value of Cu . Reconciling this Cu -range mismatch between the two derivation routes (perturbative velocity series vs. the closed-form pressure-gradient inversion) is identified as the key open question for extending this model, and is discussed further below.

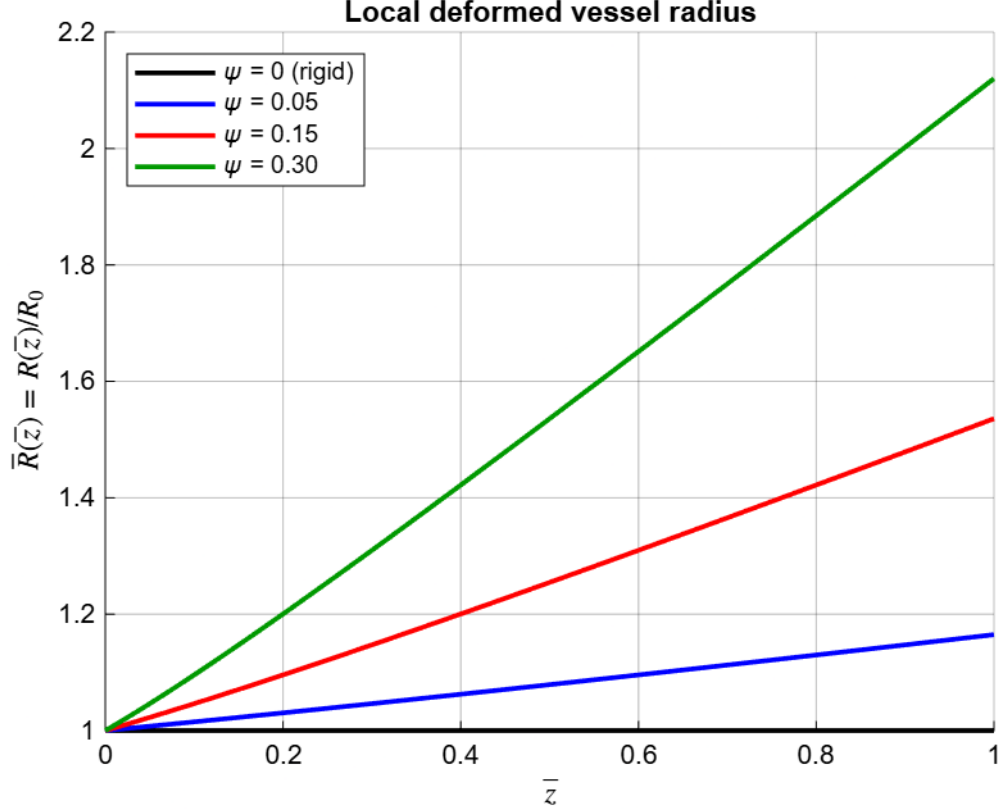


Figure 5: Velocity profile and interface continuity.

7.5 Comparison with Chebbi (2015) and conclusion

Overall, the results both reproduce and extend the rigid-tube Fåhræus–Lindqvist framework of Chebbi [6]: the $\psi = 0$ limit matches the constant pressure-gradient, fixed- σ behavior of the original marginal-zone model, while the $\psi > 0$ cases reveal a previously unmodeled coupling between wall compliance, axial pressure build-up, and core hematocrit depletion. Where Chebbi’s model and its antecedents (Fournier [3]; Sharan & Popel [17]) treat the vessel wall as a passive, unyielding boundary, the present formulation shows that physiologically realistic wall compliance amplifies the apparent Fåhræus effect by up to a factor of two over the length of a single microvessel segment, with corresponding double-digit-percentage increases in the pressure drop needed to sustain flow. This suggests that apparent-viscosity correlations calibrated on rigid-tube assumptions may systematically underpredict the true pressure demand in compliant microvasculature (e.g., venules, post-capillary vessels, or PDMS-based lab-on-a-chip channels [1, 9]), motivating the deformable-wall extension presented in this work.

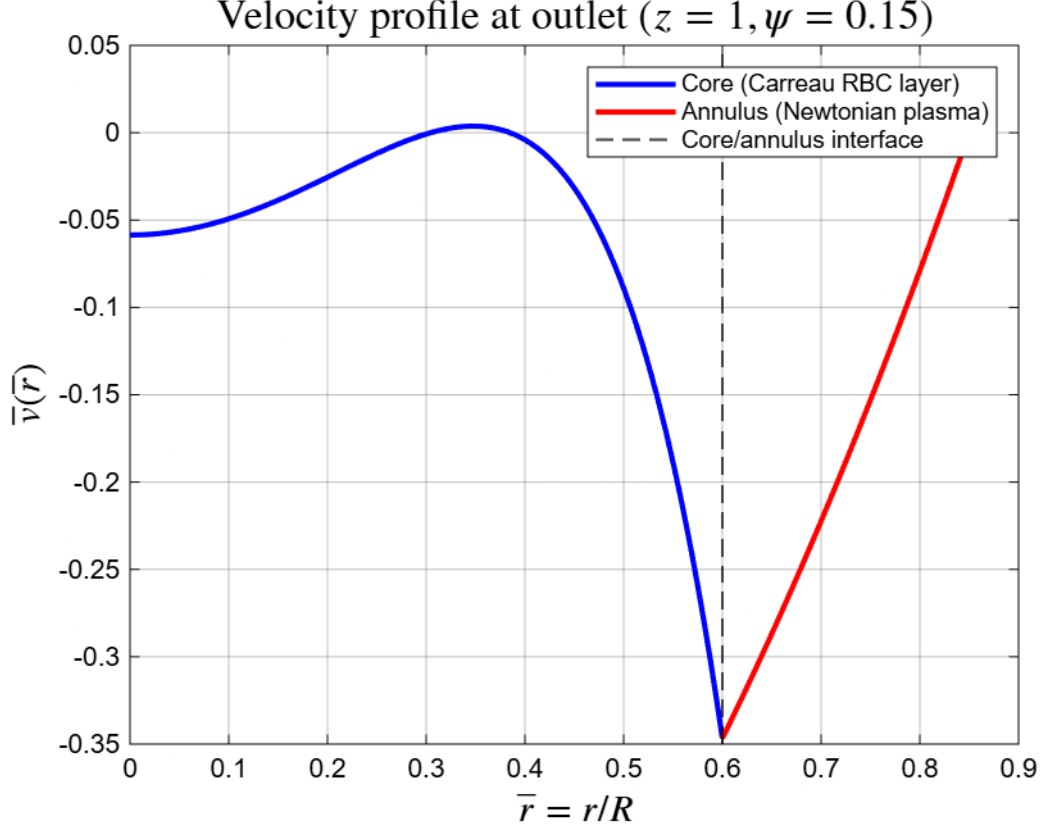


Figure 6: Comparison with existing rigid-tube frameworks.

8 Methodology

All results were generated using a single MATLAB script (`blood_flow_solver.m`) implementing the following pipeline:

1. **Lumped-constant evaluation.** The geometric/rheological coefficients A_1 , B_1 , B_3 , and B_5 were computed once from the fixed parameters ($H_D = 0.45$, $r_1^*/R_0 = 0.60$, $r_2^*/R_0 = 0.25$, $\mu_r = 1.0$, $\beta = 0.10$, $n = 0.60$, $Cu = 2.0$), since these do not depend on $\bar{z}$ or $\bar{p}$.
2. **ODE right-hand side.** A nested function evaluates $d\bar{p}/d\bar{z} = x(\bar{p})$ at any given $\bar{p}$ by (i) computing $\sigma(\bar{p})$ and $F(\bar{p})$ from the relevant governing equations, (ii) forming $M(\bar{p}) = 1 - F/\sigma^2$, and (iii) solving the resulting quadratic-in- x^2 equation for the physically admissible, real, non-negative root, with a numerically guarded fallback for the near-singular case $M \rightarrow 0$.
3. **Integration.** `ode45` integrates this first-order autonomous ODE from $\bar{z} = 0$ (with $\bar{p}(0) = 0$) to $\bar{z} = 1$, for four values of the wall-compliance parameter $\psi \in \{0, 0.05, 0.15, 0.30\}$ using tight tolerances ($\text{RelTol}=1\text{e-}10$, $\text{AbsTol}=1\text{e-}13$) to ensure the plotted curves are converged rather than artifacts of coarse stepping.
4. **Post-processing.** At each solution point, $\sigma(\bar{z})$ and $\bar{R}(\bar{z})$ are recomputed directly from the converged $\bar{p}(\bar{z})$ trajectory, producing the relevant figures without any additional integration.

5. **Velocity reconstruction.** At the outlet station of the $\psi = 0.15$ run, the converged $d\bar{p}/d\bar{z}$ value was substituted into the closed-form core and annulus velocity expressions to generate the final velocity profile figure, providing an independent numerical check of interface continuity.

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